

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall
Programme: BE
Course: Image Processing Pattern Recognition

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

Define Digital Image Processing. Also discuss the various fundamental steps in DIP with block diagram. 7

Define linear stretching. Equalize the given histogram and also draw the modification of it. 8

Gray level	0	1	2	3	4	5	6	7
No. of pixel	750	1020	850	660	340	240	123	80

- What do you mean by Image Enhancement in spatial domain? 7
- Compare low pass filter with high pass filter with example! 7
- How can eliminate the blurring from the image using the of noise model restoration techniques? 8
- What is Hadamard Transform? Explain its properties. 7
- Define Redundancy. What are image compression techniques? Source generate the symbol s_1, s_2, s_3, s_4, s_5 randomly with probability $p_1=0.4, p_2=0.2, p_3=0.2, p_4=0.1$ and $p_5=0.1$ respectively. Generate the codeword for each symbol using Huffman coding. Also calculate Entropy of the system. 8
- a) Discuss the importance of image compression. Explain lossless Predictive coding with block diagram 7
- b) Explain Dilation and Erosion operations in Morphological Image Processing with suitable example. 8

OR

Define point operation. A 4×4 size image has following intensity information.

1	2	3	4
5	6	7	8
9	8	6	7
5	4	3	2

Change this image into 8×8 size image using interpolation method.

5. a) Compare the Image Segmentation by local thresholding and basic global thresholding detail.
- b) Explain with example the growing and merging techniques for image segmentation
6. a) Explain chain code and shape number for image representation
- b) What is Neural Network? Explain the process of single layer perceptron.
7. Write short notes on: (Any two)
 - a) Pattern Recognition
 - b) Fourier Descriptor
 - c) FFT

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2015

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Draw the diagram of Digital Image Processing system and explain all the steps. 7
- b) The focal length of the human eye is 17 mm when it focuses on an object farther away than about 3m. Now, find out the size of the image formed in the fovea of a person who focuses on a 15 m tall tree that is 100 m away. 8
- a) Describe in brief, the working of sampling and quantization in digitizing an analog image. 7
- b) What are the characteristics of logarithmic and inverse log transformation? Exemplify with necessary diagrams. 8
- a) What is a 1-D FFT? Describe its properties. Also explain its computational advantage over DFT. 7
- b) Obtain the Huffman codes from the image given below and also compute the efficiency of the coding compared to natural binary bits representation of the image. 8

Gray Level	0	1	2	3	4	5	6	7
Frequency	200	150	75	100	50	125	50	250

- a) What is the fundamental difference between Lossy and Lossless predictive coding techniques? Explain with their block diagrams. 7
- b) Define pattern and pattern classes. Prove with an example that the accuracy of classification depends heavily on the selection of descriptors. 8
- a) What are gray level features? State types of edges and their possible derivative filters for the detection. 7
- b) What are line, spot, and edge in a digital image? Explain the method of detecting lines using compass detectors. 8

6.
 - a) Write down basic formulations of region based segmentation. Also describe the technique of region splitting and merging technique.
 - b) Describe a technique of image segmentation by histogram based thresholding.
7. Write short notes on: (Any two)
 - a) Bit extraction
 - b) Laplacian filter for edge detection
 - c) Hopfield Networks

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2016

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) What is a digital image? Explain the relationship between pixels with their adjacency in image processing. 7

- b) Compute the Histogram of Equalization from the given data. 8

r_k	0	1	2	3	4	5	6	7
n_k	5320	1000	500	525	1236	956	856	128

- a) What do you mean by Image Sharpening? Explain Median Filter with suitable example. 7

- b) Calculate the Haar transform T from the given image matrix F. 8

F=

1	0	0	1
1	1	0	1
1	0	1	0
1	0	1	1

And then reconstruct the original F by performing inverse Haar transform on T.

- a) Explain basic steps of filtering in Fourier domain with necessary figures. 7

- b) What is Noise in Image? Explain the Adaptive Median Filter in detail. 8

- a) Why is image compression necessary? Explain lossy predictive coding with examples. 7

- b) What is Coding Redundancy? Construct Huffman code for each gray level and calculate average bit length from the given data. 8

r	0	1	2	3	4	5	6	7
N(r)	20	30	60	120	130	160	204	300

Where, r = Gray level

$N(r)$ = Pixels having r^{th} gray level

- a) Given an image A and a structuring element B below, calculate erosion $A \ominus B$, dilation $A \oplus B$, opening $A \circ B$ and closing $A \bullet B$.

- b) Given, $A =$

0	0	0	0	0	0	0	0
0	0	0	0	0	1	1	0
0	1	1	1	0	1	1	0
0	1	1	1	0	1	1	0
0	1	1	1	0	0	0	0
0	1	1	1	0	0	0	0
0	0	0	0	0	0	0	0

And $B =$

0	1	0
1	1	1
0	1	0

6. a) What is Image Segmentation? Compare Prewitt filter with Sobel filter with their equations, matrices and characteristics.
- b) What is the role of Edge Linking in boundary detection? Explain the Hough Transform for detecting lines with suitable figures.
7. Write short notes on: (Any two)
- a) Chain codes
- b) Perceptron training algorithm
- c) Pattern and Pattern class

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2017

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) What is digital image? How is it represented? Explain briefly about the elements used in image processing? 8
- b) What do you mean by distance measures in image processing system? Explain with types. 7
- a) Define Point Operation. Explain the technique used by Bit plane intensity level slicing for the purpose of image enhancement. 8
- b) What is histogram? What is the importance of histogram modeling in image processing? Given the following gray level of an image. Compute the gray levels after histogram equalization. 7
- | | | | | | | | | |
|------------|-----|------|-----|----|-----|-----|-----|----|
| Grey Level | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Frequency | 790 | 1024 | 850 | 65 | 329 | 245 | 122 | 81 |
- a) Differentiate between Spatial domain enhancement and Frequency domain enhancement. Mention the properties of Fourier transform 7
- b) Explain different technique used to sharpen an image. Explain Hadamard Transform. 8
- a) Compare different types of Noise in terms of Spatial and frequency Properties. Suggest Corresponding filtering techniques for each type of Noise. 8
- b) Explain the basic concept of differential predictive coding in terms of lossless coding with suitable figure. 7
- a) Define Dilation, Erosion, Opening and Closing. Choose different objects and structuring elements to show the result of these operations. 8
- b) What is region based segmentation? Explain the types of region segmentation. 7

6.
 - a) Explain edge detection techniques with its assumption. Explain Hough transform for detecting lines with suitable figures.
 - b) What is pattern recognition? How does pattern/image is recognized by a computer?
7. Write short notes on: (Any two)
 - a) Hopfield Network
 - b) Run Length Coding
 - c) Shape number and chain codes

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year : 2017
Programme: BE		Full Marks: 100
Course: Image Processing and Pattern Recognition		Pass Marks: 45
		Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) What are the components of Digital Image Processing? Explain the application of it in the field of medical science. 8
- b) What is the basic of spatial filtering? Compare Mean Filter with Median Filter along with suitable example. 7
- a) What is Frequency Domain? Explain the equations of 2D-DFT along with their properties. 7
- b) Compare point operation with neighborhood operation in image processing. Explain the process of digital negative. 8
- a) What is sharpening in frequency domain? Compare Butterworth high pass filter with Gaussian high pass filter with equations. 7
- b) What is Periodic Noise? Explain any two Band Reject Filters to remove periodic noise in frequency domain. 8
- a) What is the purpose of compression in image processing? Compare fixed length coding with variable length coding along with example. 8
- b) What is Psychovisual Redundancy? Explain lossy predictive coding with require equation and block diagram. 7
- a) What is similarity based segmentation? Explain the process of image segmentation by region splitting and merging. 7
- b) Compare and explain Dilation and Erosion in image processing with necessary equations and suitable figures. 8
- a) What is the role of Gradient Operator? Compare and explain the Robert-cross operator, Prewitt operator and Sobel operator. 7
- b) What can we Represent the Regions in Image? Compare and explain the process of 4-directional chain code with 8-directional chain code. 8

Write short notes on: (Any two)

2×5

- a) Matching by Correlation
- b) Pattern Recognition Vs Image Processing
- c) Basic Adaptive Thresholding

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Image Processing and Pattern Recognition

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) What is a digital Image? Explain the fundamental steps in Digital Image Processing. 7

- b) Compute the histogram equalization from the given data. 8

r_k	0	1	2	3	4	5	6	7
n_k	5320	1000	500	525	1236	956	128	856

- a) What is zooming? Illustrate with an example the concept of zooming by interpolation and replication. 7

- b) What are the basic steps for filtering an image in frequency domain? Elaborate with examples. 8

- a) Calculate Haar transform T from given image matrix F. 7
F=

1	0	0	1
1	1	0	1
1	0	1	0
1	0	1	1

- b) Explain the noise degradation and restoration model with necessary diagram in detail. 8

- a) Define Redundancy. Suppose a source generates the symbols s_1, s_2, s_3, s_4, s_5 randomly with probability $p_1=0.4, p_2=0.2, p_3=0.2, p_4=0.1$ and $p_5=0.1$ respectively. Generate the code-word for each symbol using Huffman coding and also calculate entropy and efficiency. 8

- b) Explain Lossy Predictive Coding along with required equations and suitable block diagram. 7

- a) Compare and explain the process of Dilation and Erosion in image processing with necessary equations and suitable figures. 8

- b) Explain image segmentation by Threshold method.
- 6. a) Explain the region growing technique for image segmentation. What are the problems associated with it?
- b) What is neural network? Explain how it can be used for pattern recognition.
- 7. Write short notes on: (Any two)
 - a) Pattern recognition system
 - b) Shape number
 - c) Decision Theoretic Method

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2018

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define Digital Image Processing. Also discuss the various fundamental steps in DIP with block diagram. 8
- b) Discuss the importance of histogram modelling. Explain the algorithm for histogram equalization. 7
- a) Explain the term intensity level slicing and bit plane slicing. Explain adaptive median filter with suitable algorithm for its implementation. 7
- b) Write expression for forward and inverse Discrete Fourier Transform (DFT) for 2D signal. What are the properties of DFT.? 8
- a) Define Hadamard Transform and derive the 8x8 transform matrix for Hadamard Transform. 8
- b) Explain about the noise restoration model. How the periodic noise occurring in an image can be removed? 7
- a) Define Redundancy. What are image compression techniques? Source generate the symbol s1, s2, s3, s4, s5 randomly with probability $p_1=0.4$, $p_2=0.2$, $p_3=0.2$, $p_4=0.1$ and $p_5=0.1$ respectively. Generate the codeword for each symbol using Huffman coding. 8
- b) Discuss the importance of image compression. Explain lossless Predictive coding with block diagram 7
- a) With necessary figures, explain the opening and closing. 8
- b) How can you detect edges with gradient filters? Give different first order derivative based (gradient) filters. 7

6.
 - a) What do you understand by image segmentation? How discontinuity based segmentation approach is performed in image? Explain.
 - b) What is pattern and pattern classes? How is an object recognized by minimum distance classifier?
7. Write short notes on: (Any two)
 - a) Pattern Recognition
 - b) Fourier Descriptor
 - c) Bayesian Classifier

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2019

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define digital image processing? How is an image transformed into Grey level image? Explain with an example and pseudo code. 8
- b) What is zooming? Illustrate with an example by interpolation and replication. 7
- a) Compute the histogram equalization from the given data. 8

Input Image :- 4x4 Image

Gray Scale=[0,9]

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

- b) Write algorithm of Haar Transform. Compute the Haar Transformation for the order N=2. 7
- a) What is histogram equalization? Write the algorithm and pseudo code for histogram equalization with suitable example. 8
- b) List out some important noise probability density functions used in image processing and sketch their plots. 7
- a) What do you mean by predictive coding in image compression? 8
- b) With necessary figures, explain the opening and closing. Briefly explain the algorithm for lossy predictive coding using feedback. 7
- a) Construct Huffman code for each gray level and find the compression ratio and coding efficiency. 8

Gray Level	0	1	2	3	4	5	6	7
No. of pixels	113	139	142	145	181	105	50	1323

- b) Explain Segmentation by thresholding with its feature thresholding and Amplitude thresholding.
6. a) How can you detect edges with gradient filters? Give different first order derivative based (gradient) filters.
- b) Given an image, "A" represents its pixel position. If A* is the starting pixel, write down the 8-chain code and find shape number of it.

			A*	A		
		A			A	
		A		A		
	A				A	
		A	A	A		

7. Write short notes on: (Any two)
- a) Neural Network
- b) Learning in perception network
- c) Dilation and Erosion

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2019

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define digital image. Explain the process of image sampling and quantization in the digital image representation with figures. 8
- b) Write the algorithm and pseudo code for average filtering and apply 2D Average filtering on the given image. Use 3×3 sampling window and Keep border values unchanged. 7

1	4	0	1	3	1
2	2	4	2	2	3
1	0	1	0	1	0
1	2	1	0	2	2
2	5	3	1	2	5
1	1	4	2	5	0

2. a) What is histograms equalization? Write the algorithm and pseudo code for histogram equalization with suitable example. 7
- b) Compute the 2-D discrete Walsh-Hadamard transform of the given image block below: 8

5	6	8	10
6	6	5	7
4	5	3	6
7	8	3	5

3. a) What are the usages of derivative based filters. Derive mask for Laplacian second order derivative based.
- b) What is Image Restoration? Compare it with image enhancement.
4. a) Construct Huffman code for each gray level.

Gray (r)	0	1	2	3	4	5	6	7
n(r)	60	100	260	20	150	90	110	234

Where, r = Gray level

$n(r)$ = No. of Pixels having r^{th} gray level

- b) Explain the working principles of lossy predictive coding with necessary block diagram.
5. a) What is morphological image processing? With necessary figures, explain the opening and closing.
- a) How can you detect edges with gradient filters? Give different first order derivative based (gradient) filters.
6. a) What do you mean by image segmentation, give the algorithm for basic adaptive thresholding.
- b) Explain Minimum Distance Classifier for Pattern Classification with necessary equations and an example.
7. Write short notes on: (Any two)
 - a) Shape Numbers
 - b) Pattern with Pattern Class
 - c) Gray level slicing

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2020

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) What is a digital image? Explain spatial resolution and intensity level resolution with examples. 7
- b) State different types of Gray Level Transformations. Explain any three of them in brief. 8
- a) Given below are gray level frequencies of a 3 bit image. Equalize the histogram over the range. 7

Gray Level (I)	0	1	2	3	4	5	6	7
Frequency n(I)	32	160	128	512	192	256	704	64

- b) Perform Haar transform and its Inverse for 4x4 image given below. 8

1	2	5	0
7	2	5	6
4	5	3	3
7	6	1	2

- a) What are the steps involved in frequency domain filtering? Explain the essential properties of 2-D Discrete Fourier Transform? 7
- b) Compare image enhancement and image restoration. Explain different noise models in an image with their probability density functions. 8
- a) Construct Huffman coding for each gray level and find compression ratio and coding efficiency: 8

Gray level(r)	0	1	2	3	4	5	6	7
Frequency N_r	500	200	250	100	250	500	150	50

- b) Differentiate between lossless and Lossy Compressions.
5. a) Find Dilation and Erosion for the following image A and structuring element B.

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

A

1
1
1

B

- b) Explain basic global thresholding and region splitting in image segmentation with example. 8
6. a) Describe Edge Detection method for Image Segmentation 7
- b) Define Chain codes. Explain 4 and 8-connectivity segments to explain chain codes with suitable examples. 8
7. Write short notes on: (Any two) 2
- a) Neural Network in Image Processing
- b) Pattern recognition in Image
- c) Spatial low pass filter.

POKHARA UNIVERSITY

Level: Bachelor

Semester – Spring

Year: 2020

Program: BE

Full Marks: 70

Course: Image Processing and Pattern Recognition

Pass Marks: 31.5

Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

Group - A: (5×10=50)

- Q. N. 1 Consider the following image with the new pixel at (2,2) if the smoothing is done using a 3×3 bit neighbourhood finding. 10

3	4	1	7	7
2	5	6	2	0
1	5	7	6	4
1	2	2	3	0
4	7	5	6	7

- a) Mean Filter
- b) Weighted average filter
- c) Median Filter
- d) Min Filter
- e) Max Filter

- Q. N. 2 Explain the components of digital image processing. How many images of size 1920×1720 with 8-bit gray value can be stored in a 512 MB storage space? 10

OR

Compare between Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT) with reference to Digital Image Processing. Derive the equations of FFT algorithm for one dimensional case.

- Q. N. 3 Using the concept of Hit, Fit and Miss, and taking a structural element of your choice, explain erosion and dilation with pictorial examples. 10

- Q. N. 4 "Edge linking is used in boundary detection of an image? Justify. Explain the method of edge linking using Hough Transform in detail with suitable figures. 3+7

- Q. N. 5 Differentiate between lossless & lossy compression? Consider the source with 8 message to design Huffman code and efficiency for intensity values in following normalized histogram. 4+6

Gray Level (r_k)	0	1	2	3	4	5	6
$P(r_k)$	0.25	0.25	0.125	0.125	0.125	0.0625	0.0625

Group - B: (1×20=20)

- Q. N. 6**
- a) What is time and frequency domain transformation?
 - b) Compute the DFT of $f(x) = \{1, -1, 3, 2\}$
 - c) How do you convert spatial domain into frequency domain?
 - d) Apply contrast stretching technique on 3-bit gray level image of size 4x4.

6	6	2	3
6	5	4	7
2	2	4	1
2	3	6	4

2
8
4
6

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Year : 2021
Full Marks: 100
Pass Marks: 45
Course: Image Processing and Pattern Recognition
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) What is brightness adaptation and discrimination? Explain the various key stages in digital image processing with essential features associated with each. 7
- b) What is zooming? Illustrate with an example by interpolation and replication. 8
- a) What are the uses of histogram processing? Perform the histogram equalization on the following 8-bit gray level image and also scale the intensity from 1 to 20. 8

3	2	4	5
1	2	1	2
7	3	1	2
7	6	4	7

- b) What are the steps involved in frequency domain filtering? Explain the essential properties of 2-D Discrete Fourier Transform. 7

- a) Write algorithm of Haar Transform. Compute the Haar Transformation for the order $N=2$. 7

- b) Compare image enhancement and image restoration. Explain different noise models in an image with their probability density functions? 8

- a) Define psychovisual redundancy. Find Huffman code and calculate the coding efficiency of the following 6×6 image: 8

254	254	100	100	100	12
254	254	100	100	100	12
254	254	100	100	100	12
254	254	100	100	100	12
254	254	100	100	100	12
254	254	100	100	100	12

50	50	50	50	50	50
254	254	100	100	12	12
254	254	100	100	12	12
254	254	100	100	12	12

- b) Define structuring elements, dilation and erosion in morphological image processing. Explain opening and closing with suitable examples.
5. a) Explain basic global thresholding and region splitting in image segmentation with essential pseudocode and algorithms.
- b) Define image segmentation. What are the problems associated with the first and second order derivative filters for edge detection during segmentation? Explain with suitable solutions.
6. a) Given an image, "A" represents its pixel position. If A* is the starting pixel, write down the 8-chain code and find shape number of it.

			A*	A		
		A			A	
		A		A		
	A				A	
		A	A	A		

- b) How can we use various neural network techniques for object recognition in image processing? Explain.
7. Write short notes on: (Any two)
- a) Steps of Pattern recognition system
- b) Lossy image compression
- c) Discrete Cosine Transform

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2021

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Explain basic relationships like- neighbours, connectivity and distance measures between pixels in any digital image with suitable examples 7
- b) Define intensity level slicing. How can you use histogram equalization technique for contrast enhancement? Explain with suitable example. 8
- a) Consider the following image with the new pixel at (2,2) if the smoothing is done using a 3x3 bit neighbourhood. Find 8
- i. Mean Filter
 - ii. Weighted average filter
 - iii. Median Filter
 - iv. Min Filter
 - v. Max Filter

1	2	1	3	8
3	8	8	5	1
5	7	5	4	4
2	2	1	8	6
6	5	3	3	6

- b) Explain Haar Transform with its algorithm for the order $N=2$. 7
3. a) What are the main sources of noises in digital image? How can you use harmonic and contra-harmonic mean filters to remove noise in an image? 8

- b) Define structuring elements. Explain opening and closing in morphological image processing with suitable examples.
- 4. a) Define psychovisual redundancy. Employ Huffman Coding algorithm to determine the bits required for encoding the message 'mississippi'.
b) List out the different methods of selecting Threshold value for Image Segmentation. Explain any two of them in brief.
- 5. a) Explain region splitting and region growing in image segmentation with essential pseudocode and algorithms.
b) Explain, how you can use Hamming neural network for object recognition. Explain the process of single layer perceptron.
- 6. a) Define Chain codes. Explain 4 and 8-connectivity segments to explain chain codes with suitable examples.
b) What is pattern recognition? Briefly explain the various steps of pattern recognition.
- 7. Write short notes on: (Any two)
 - a) Discrete Cosine Transform
 - b) Differentiate lossless and Lossy Compressions.
 - c) Periodic Noise Reduction

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2022

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define Digital Image. Digital Image Processing and Image Acquisition. Differentiate between Image Sampling and Image quantization. 4+4
- b) Apply contrast stretching technique on following 3-bit grey level image of size. 7

2	1	2	1
4	5	5	6
3	2	1	4
6	2	1	6

2. a) Explain zooming by replication and interpolation for the magnification of an image. 4+4
- b) Explain Hadamard Transform with its algorithm for the order $N=4$. 7
3. a) Complete DFT of $f(x) = \{1, -1, -2, 2\}$ 8
- b) Define salt and pepper noise. Explain the Adaptive mean Filter. 7
4. a) Find the average length of following 3-bit image after using Huffman Coding technique. 8

r_k	0	1	2	3	4	5	6	7
count	113	139	142	145	181	105	52	1323

- b) Differentiate lossless and lossy Compressions. 7
5. a) Find closing and opening of following image(A) for the given structuring element(B). 8

A=

1	0	0	0	1
0	1	0	1	0
0	0	1	0	0
0	1	0	1	0
1	0	0	0	1

B=

1	1*	1
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- b) Explain Hough Transform.
6. a) Explain region splitting and region growing in image segmentation with an algorithm.
b) Define Chain codes, signature and shape number. Explain their use with suitable examples.
7. Write short notes on: (Any two)
- a) Run Length Coding
 - b) Pattern and Pattern Classes
 - c) Neural Networks for image pattern recognition

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2023

Programme: BE

Full Marks: 100

Course: Image Processing and Pattern Recognition

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define neighboring of pixels. Explain its types with example of each types. 7

- b) Compute the Histogram of Equalization of given data. 8

Grey level	0	1	2	3	4	5	6	7
No of Pixels	5320	1000	500	525	1236	956	856	128

- a) What is zooming? How replication and Interpolation techniques be used for zoom? Explain with a suitable example. 7

- b) Describe the method of image sharpening using the Laplacian. 8

- a) When do we need to center the Fourier transform of an image? Prove that multiplying the image intensities by $(-1)^{x+y}$, where (x, y) represents the coordinate of the pixels, centers the Fourier transform. 7

- b) Define salt-and-pepper noise. Which spatial filter is used to reduce salt-and-pepper noise? Why? 8

- a) Why is image compression necessary? Explain lossy predictive coding with examples. 7

- b) Find Huffman codes from given histogram information of an image. 8

Grey Level	0	1	2	3	4	5	6	7
Frequency	35	30	15	10	38	80	38	0

- a) Define the term discontinuity? Explain point, line and edge detection methods. 8

- b) Define the morphological erosion and dilation operators. Which of these can be considered as a noise-reduction technique? Why? 7

6.
 - a) Describe how a grayscale image can be segmented using intensity thresholding. You need to explain the cases of single and multiple thresholds.
 - b) What is Neural Network? Explain the application of neural network in pattern recognition.
7. Write short notes on: (Any two)
 - a) Fourier descriptors
 - b) Steps in pattern recognition
 - c) Image Averaging

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Image Processing and Pattern Recognitions

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define digital image processing. Explain briefly about Components of an Image Processing System. 8
- b) Apply contrast stretching technique on following 3-bit grey level image of size 4×4 . 7

4	1	2	1
7	5	6	5
3	2	2	4
2	2	3	6

2. a) Differentiate between smoothening and sharpening filters. Explain some basic grey level transformations techniques used for image enhancement in spatial domain. 7
- b) Compute the 2-D discrete Walsh-Hadamard transform of the given image block below: 8

5	6	8	10
6	6	6	7
3	5	3	6
7	8	6	7

3. a) Why do we use Haar Transform in Image processing? Explain its algorithm for the order $N=2$. 7
- b) Define salt-and-pepper noise. Which spatial filter is used to reduce salt-and-pepper noise? Why? 8

4. a) What do you mean by lossless image compression? Briefly explain the algorithm for run length coding with suitable example. 8
- b) Find compression ratio and coding efficiency of following 3-bit image after using Huffman Coding technique. 7

Gray level	0	1	2	3	4	5	6	7
No. of Pixel	30	35	38	10	15	10	38	80

5. a) Define the morphological erosion and dilation operators. Which of these can be considered as a noise-reduction technique? Why? 7
- b) Describe how a grayscale image can be segmented using intensity thresholding. You need to explain the cases of single and multiple thresholds. 8
6. a) How can you detect edges with gradient filters? Give different first order derivative based (gradient) filters. 7
- b) For the following image, write down the 8-chain code and find the shape number. I* is the starting point and the rotation is clock-wise direction. 8

		I*	I	I			
	I				I		
	I					I	
	I					I	
		I			I		
			I	I			

7. Write short notes on: (Any two) 2×5
- a) Neural Network
- b) Steps in Pattern Recognition
- c) Image Restoration Vs Image Enhancement

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Image Processing and Pattern Recognition

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Fundamental steps in Digital Image Processing. 8
b) Define Image Enhancement. Describe Contrast stretching and Thresholding in the field of Image Enhancement. 7
2. a) Gray Level histogram of an image represented in 3 bit system is given below 8

Gray level	0	1	2	3	4	5	6	7
Frequency	0	0	50	60	50	20	10	0

Stretch the contrast of histogram over the entire range.

b) Why do we use Haar Transform in Image processing? Explain its algorithm for the order $N=2$. 7
3. a) When do we need to center the Fourier transform of an image? Prove that multiplying the image intensities by $(-1)^{x+y}$, where (x,y) represents the coordinate of the pixels, centers the Fourier transform. 7
b) Define salt-and-pepper noise. Which spatial filter is used to reduce salt-and-pepper noise? Why? 8
4. a) What is coding redundancy? Construct Huffman code for each gray level. 8

M	0	1	2	3	4	5	6	7
NM	41	101	300	10	163	75	124	210

where, M = Gray level
 NM = Pixels having M^{th} gray level

b) Differentiate lossless and lossy Compressions with the areas of their uses. 7
5. a) Define the morphological erosion and dilation operators. Which of these can be considered as a noise-reduction technique? Why? 7

b) Describe the region merging and region splitting technique for image segmentation with an example. 8

a) Explain Hough Transform. 7

b) Define Chain codes, signature and shape number. Explain their use with suitable examples. 8

Write short notes on: (Any two) 2×5

a) Patterns and Patterns Classes

b) Neural Network

c) Steps in Pattern Recognition